

Activity 14 Continuous Improvement with GenAI - PDCA and DMAIC

Activity 14 | Topic 02 | C013 | Version v1.0

Course: Generative AI for Project Management (C013) Topic 02: Project Execution and Delivery with GenAI Learning outcome: LO2 - Structure an improvement cycle and control the result. Tools: GenAI tool | Plan Do Check Act.pptx | DMAIC.pptx

Goal

Having found a root cause, you need a disciplined way to fix it and make the fix stick. You will run the same improvement through both PDCA and DMAIC and see where they differ.

What you'll produce

A completed PDCA cycle and a DMAIC project definition for your improvement.

Activity files

- [Activity guide PDF](./activity-guide.pdf)
- [Prompt sheet](./prompts.pdf)
- [Detailed instruction manual](./instruction-manual.pdf)
- [Mock data workbook](./mock-data.xlsx)

Core prompt

Design a PDCA cycle to address [root cause]. For Plan, state the evidence and prioritised solution; for

Step-by-step

1. Take the root cause you identified in Activity 13. Open 'Plan Do Check Act.pptx'.
2. Prompt GenAI: 'Design a PDCA cycle to address [root cause]. For Plan, state the data reviewed and the prioritised solutions; for Do, define a small pilot; for Check, define the measure of success; for Act, state what happens if it works and if it does not.'
3. Sharpen the Check stage - this is where AI drafts are weakest. Insist on a specific number and a date: 'rework hours per sprint below 8 by 30 September', not 'improved efficiency'.
4. Complete the PDCA template. Confirm your Do stage is genuinely a small pilot rather than a full rollout.
5. Now open 'DMAIC.pptx' and prompt: 'Structure the same improvement as a DMAIC project. Define the gap, state how I would measure the current problem with real data, analyse root cause, propose the improvement, and specify the control that keeps the process in place.'
6. Compare the two: PDCA is a fast learning loop; DMAIC is a heavier, data-driven method for a chronic problem. State which fits your improvement and why.
7. Pay attention to the Control stage - ask GenAI 'what would cause this improvement to decay after six months?' and build a control against it.
8. Share your improvement plan with the class.

Check your work

Check: Your Check/Control stages specify a number and a date, your Do stage is a pilot, and you can justify choosing PDCA or DMAIC for this problem.

Remember: GenAI gives you a fast first draft. Check every figure, source and assumption before the output leaves your hands - you remain accountable for it.

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